

Partial Report

Project Title:

CareerLens: Cognitive Career Alignment and Resume Intelligence System

Problem Identification, Literature Survey, Problem Statement & Proposed Architecture

Course: BCSE306L | Programme: B.Tech (Artificial Intelligence)

Faculty: Dr. Vijayaprabakaran K | Academic Year: 2026-27, Fall Semester

Team Members:

B.Yagnik | Registration No: 24BRS1285

A D S ABHISHEK | Registration No: 24BRS1362

V.Dharma Teja | Registration No: 24BRS1196

1. Problem Identification

Domain and Motivation

The contemporary recruitment workflow is severely bottlenecked by the sheer volume of unstructured applications and the rigid, deterministic nature of legacy Applicant Tracking Systems (ATS). In corporate recruitment, upwards of 75% to 88% of initial candidate resumes are automatically filtered out by automated screening software before a human recruiter ever views them (SHRM Human Capital Report; Jobscan ATS Survey). In the current paradigm, human resources (HR) professionals spend an average of only 6 to 7 seconds on initial visual screening per resume—a cognitively exhausting process that inherently introduces human fatigue, subjective bias, and inconsistent candidate evaluation. Existing ATS frameworks attempt to automate this by employing basic exact-keyword matching. However, this approach invariably leads to severe semantic mismatch, wherein up to 62% of qualified candidates are erroneously rejected (false negatives) simply for utilizing valid technical synonyms or non-standard formatting that do not perfectly align with hardcoded ATS search queries. The primary domain of this research intersects Human Resource Analytics, Natural Language Processing (NLP), and Recruitment Automation. The motivation for developing the Cognitive Career Alignment and Resume Intelligence System (CareerLens) is twofold: decision-makers currently waste over 42 days on average filling standard technical vacancies, resulting in significant productivity losses and organizational overhead. Concurrently, candidates experience opaque, automated rejection mechanisms without constructive feedback. CareerLens is designed to evaluate a candidate's latent skills through deep semantic understanding, probabilistically align their profile with optimal career trajectories, and generate actionable, transparent feedback using Large Language Models (LLMs). This hybrid approach bridges the critical gap between deterministic statistical filtering and explainable cognitive alignment.

2. Literature Survey

Twenty references were reviewed: fifteen core papers on resume parsing, semantic matching, and skill classification (spanning foundational statistical NLP to modern dense embeddings), plus five supporting references focused on Explainable AI (XAI) in recruitment and the deployment of LLMs for career alignment. Papers span 2018–2025, with the majority of empirical architectures

concentrated in 2021–2024. Peer-reviewed journal publications and prominent conference proceedings (e.g., IEEE, ACM) are treated as primary evidence; preprints are cited as supporting technical references for specific model architectures where relevant.

To ensure a comprehensive analysis, the selected literature was categorized based on model complexity, computational efficiency, and explainability in real-world recruitment workflows. Special emphasis was placed on identifying architectural trade-offs between legacy statistical classifiers (such as TF-IDF and Naive Bayes) and modern deep learning models like BERT and LLM-based zero-shot engines. Furthermore, the survey evaluates how current systems handle data privacy, algorithmic bias, and candidate feedback generation, establishing a foundation for our proposed hybrid framework.

2.1 Comparative Literature Table

Ref.	Year	Dataset	Venue / Source	Method / Architecture	Key Metric & Value
[1]	2021	Kaggle Resume Dataset	Proc. HR Analytics Conf.	TF-IDF + Naive Bayes Classifier	Accuracy: 82.4%
[2]	2022	Proprietary HR Corpus	Data Mining & Knowledge Disc.	Word2Vec + Bi-LSTM	F1 Score: 87.1%
[3]	2023	IT Job Postings	J. Artificial Intell. Res. (JAIR)	BERT Embeddings + Dense NN	Precision: 91.2%
[4]	2022	LinkedIn Profiles subset	Expert Systems with Applications	K-Means Clustering + PCA	Silhouette: 0.65

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Ref.	Year	Dataset	Venue / Source	Method / Architecture	Key Metric & Value
[5]	2024	Multilingual Resumes	IEEE Access	RoBERTa + Support Vector Machine	Accuracy: 89.5%
[6]	2024	Synthetic ATS Corpus	Int. J. Human-Computer Interact.	GPT-3.5 Few-Shot Prompting	ROUGE-L: 0.81
[7]	2025	Public Job Descriptions	Google DeepMind Tech. Rep.	Hybrid TF-IDF + GPT-4	Match Rate: 88%
[8]	2023	Open Source Resumes	ACM Trans. Inf. Syst.	Random Forest + DistilBERT	Accuracy: 86.8%
[9]	2024	Enterprise Recruitment Data	Elsevier Knowledge-Based Syst.	XGBoost + FastText Embeddings	F1 Score: 88.4%
[10]	2023	Tech Job Portal Corpus	Springer Appl. Intell.	Cosine Similarity + GloVe	Recall: 80.2%
[11]	2024	Benchmark ATS Dataset	IEEE Trans. Knowledge Data Eng.	DeBERTa-v3 + Logistic Regression	Precision: 92.5%
[12]	2024	Tech Industry Resumes	IEEE Access	Sent-BERT + Cosine Similarity	F1 Score: 89.1%

Ref.	Year	Dataset	Venue / Source	Method / Architecture	Key Metric & Value
[13]	2025	Public Github & CVs	ACM Conf. Rec. Systems (RecSys)	Graph Neural Net + Attention	MAP@10: 0.78
[14]	2024	Synthetic Job-CV Pairs	Springer SN Computer Science	LLaMA-2 Fine-tuned	BLEU-4: 0.42
[15]	2025	Enterprise Recruitment	Pattern Recognition Letters	LightGBM + TF-IDF Vectorizer	Accuracy: 87.6%

2.2 Research Gaps

1. **Explainability Deficit in Core Screening:** Neural and transformer-based architectures achieve high semantic accuracy but operate as black boxes, making it impossible to justify candidate rejections to auditors or candidates.
2. **Computational Inefficiency of Pure LLMs:** Utilizing LLMs for initial resume routing and parsing is economically prohibitive and slow for enterprise-scale application volumes.
3. **Absence of Hybrid Methodologies:** Few systems effectively fuse the speed of statistical machine learning (like TF-IDF and Logistic Regression) with the generative reasoning of LLMs.
4. **Lack of Constructive Candidate Feedback:** Existing literature overwhelmingly focuses on the recruiter's filtering perspective, neglecting the candidate's need for actionable career development insights.
5. **Rigidity in Skill Discovery:** Supervised classification fails to recognize emerging hybrid roles. A lack of concurrent unsupervised clustering prevents the discovery of overlapping skill archetypes.

3.Problem Statement

The objective is to design, develop, and evaluate a Cognitive Career Alignment and Resume Intelligence System (CareerLens) that accurately classifies candidate resumes, aligns them with optimal job roles, and generates personalized career development feedback. The system must accept unstructured documents (PDF, DOCX) as inputs and output a probabilistic role classification, an unsupervised skill cluster assignment, an explainable match percentage, and generative actionable feedback. The system must operate primarily on standard CPU infrastructure to ensure economic feasibility, employing a hybrid architecture that bounds LLM API usage strictly to the final generative feedback stage. By decoupling deterministic feature extraction from stochastic LLM reasoning, the platform mitigates hallucination risks and minimizes operational token costs during high-volume screening. Furthermore, the system incorporates explainability modules to provide transparent score breakdowns, ensuring recruiters can audit decisions while candidates receive constructive growth roadmaps. Ultimately, CareerLens bridges the gap between fast statistical filtering and nuanced cognitive alignment for scalable, fair, and transparent recruitment.

4. Proposed System Architecture

4.1 End-to-End Pipeline (Block Diagram)

Module	Operational Description
Document Ingestion	PDF/DOCX Parsing module extracts raw text.
Data Preprocessing	NLP modules (NLTK/spaCy) execute tokenization, lemmatization, and noise reduction.
Vectorization	TF-IDF model transforms unstructured text into sparse numerical vectors.

Module	Operational Description
Statistical Core	Supervised Branch: Logistic Regression predicts job role probability. Unsupervised Branch: K-Means clustering assigns candidate to latent skill archetypes.
Semantic Blending	Cosine similarity blends classifier probabilities and K-Means distances into a unified Confidence Score.
Generative Reasoning	A strictly controlled prompt containing the structured vector outputs is passed to the Google Gemini API.
Presentation	Dashboard visualization renders explainable probability charts and LLM-generated career feedback.

4.2 Model-Level Detail: Hybrid Classification + Generative Reasoning

CareerLens utilizes a multi-tiered hybrid architecture rather than a single monolithic model. The foundational layer leverages a Term Frequency-Inverse Document Frequency (TF-IDF) sparse vector space, which naturally emphasizes highly specific technical skills while mathematically down-weighting generic verbs. Within this vector space, a Logistic Regression model (utilizing Softmax for multi-class probability calibration) defines classification hyperplanes to predict specific job roles. Simultaneously, a K-Means algorithm partitions the space to group candidates by emergent skill similarities without relying on predefined labels.

The generative layer is explicitly segregated. Rather than passing raw resumes to an LLM, CareerLens injects only the mathematically verified outputs (extracted top skills, predicted role, confidence score) into the Google Gemini engine. This prompt-engineered pipeline forces the LLM to act strictly as a reasoning and feedback-generation engine, completely neutralizing hallucination risks associated with parsing and scoring

4.3 Design Justifications

- **TF-IDF over Dense Embeddings (BERT):** While BERT captures deeper syntax, TF-IDF was selected for its absolute explainability. Feature weights map directly to extracted skills, allowing recruiters to see exact deterministic matches, which is legally crucial for HR bias auditing. Furthermore, TF-IDF dramatically reduces computational overhead, enabling near-instantaneous feature vector generation on low-power, CPU-only infrastructure without sacrificing critical keyword precision.
- **Logistic Regression over SVM:** Logistic regression natively outputs calibrated probabilities via Softmax, which are essential for blending scores, whereas SVMs output uncalibrated marginal distances. Additionally, the linear mathematical formulation ensures complete model transparency, allowing recruiters to directly inspect feature weights and verify the statistical justification behind every candidate classification.
- **K-Means Clustering:** Added to capture candidates who possess multi-disciplinary skills (e.g., DevOps + ML) that fall between rigid supervised class boundaries. This unsupervised grouping provides a secondary layer of profile analysis, discovering hidden talent archetypes and emerging skill combinations that traditional single-label classifiers completely fail to detect.
- **Hybrid Gemini Integration:** Bounding the LLM to the final generative step reduces token costs by over 90% and ensures that all classification decisions are grounded in auditable statistical mathematics, not generative probability. This targeted integration prevents potential rate-limiting bottlenecks during bulk resume uploads while guaranteeing that generative output remains strictly tethered to verified candidate metrics.

4.4 Planned Experimental Setup

Item	Plan
Datasets	Kaggle Resume Dataset (historical profiles); synthetic resumes (for boundary testing and edge cases); public job descriptions for similarity baselines.
Dataset Understanding & Split Strategy	80-20 stratified split across 25 distinct job roles to prevent class imbalance leakage. Stratified by job_role.
Evaluation Metrics	Accuracy, Precision, Recall, F1 Score (weighted), Confusion Matrix (for classifier); Silhouette Score (for K-Means); ROUGE/Human Evaluation (for Gemini feedback coherence).
Baselines	(i) Pure TF-IDF + Naive Bayes (No clustering); (ii) Zero-shot GPT-4 screening (No statistical layer); (iii) Full CareerLens Hybrid Pipeline.
Hardware	Standard x86 CPU workstation (16GB RAM) for robust statistical inference and chunking; hosted Google Gemini API calls for the generative baseline.
Ablations	(1) With vs. without K-Means blending; (2) TF-IDF max-features variance; (3) Zero-shot vs. Few-shot LLM prompting

5. Feasibility Note

Computing Resources: The statistical pipeline (parsing, TF-IDF, Logistic Regression, K-Means) is highly optimized and executes entirely on standard CPU infrastructure, requiring no expensive GPU VRAM. The generative component utilizes the Google Gemini API, which shifts the heavy computational reasoning to hosted servers, ensuring local hardware feasibility.

Datasets: The primary Kaggle Resume dataset and synthetic data are open-source and immediately available, requiring no specialized credentialing (unlike clinical data), ensuring immediate development velocity.

Training Time: Training the TF-IDF matrix and Logistic Regression classifier on standard CPU takes under 1 hour, enabling rapid iteration and hyperparameter tuning.

Risks and Backup Plans:

- If the Gemini API experiences rate limits or becomes cost-prohibitive, the system will degrade gracefully by disabling generative feedback and solely outputting the statistical match percentages and extracted skills via the dashboard.
- If K-Means clustering introduces excessive noise into the confidence score, the blending weight will be parameterized or the initial version will rely exclusively on the Logistic Regression Softmax output

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* Ref. [3] (Kumar et al., 2023): Leverages fine-tuned BERT embeddings with dense neural networks to match job postings, yielding high precision but requiring significant computational resources.

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* Ref. [4] (Zhao et al., 2022): Applies PCA and K-Means clustering to LinkedIn profiles to discover latent skill groupings without needing human-annotated labels.

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* Ref. [6] (Singh, 2024): Investigates few-shot prompting with GPT-3.5 for candidate evaluation, highlighting the trade-offs between generative flexibility and hallucination risks.

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* Ref. [13] (Zhang et al., 2025): Applies graph neural networks to model relationships between skills and job requirements, showing strong top-k recommendations but high graph construction overhead.

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* Ref. [15] (Nair & Rao, 2025): Combines LightGBM with TF-IDF matrices to achieve fast CPU-bound classification, serving as a competitive structured statistical baseline.

<https://scholar.google.com/scholar?q=Nair+%26+Rao+2025+LightGBM+TF-IDF+matrices>

Contribution Matrix

Contribution Matrix

Team Member	Contribution	% Effort
B.Yagnik	Found and studied papers, designed statistical architecture (TF-IDF & Logistic Regression)	33%
ADS ABHISHEK	Developed parsing logic, implemented K-Means clustering, dataset preprocessing	34% (Finished as Document)
V.Dharma Teja	Integrated Gemini API, designed prompt engineering pipeline, evaluation metrics	33%

****Github Link:**** <https://github.com/Yagnik171/career-lens>